Mander Portman Woodward

OCR
Oxford Cambridge and RSA

Physics A

PHBx Chapters 8-10
The Physics Department

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: 45 mins

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **40**.
- The total number of marks may take into account some 'either/or' question choices.

© OCR 2017. You may photocopy this page.

Created in ExamBuilder

1. Wires **P** and **Q**, made from the same metal, are connected in **parallel** across a cell of negligible internal resistance.

The table shows some data.

Wire	**Length of wire**	**Diameter of wire**	**Mean drift velocity of electrons in the wire / mm s^{-1}**
P	L	d	0.60
Q	$3L$	$2d$	v

What is the mean drift velocity v of the electrons in wire **Q**?

A 0.15 mm s^{-1}

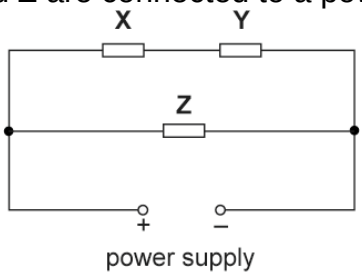
B 0.20 mm s^{-1}

C 0.30 mm s^{-1}

C 0.60 mm s^{-1}

Your answer ☐ **[1]**

2. Three identical resistors **X**, **Y** and **Z** are connected to a power supply.



The power dissipated in the resistor **Z** is 24 W.

What is the power dissipated in the resistor **Y**?

A 6.0 W

B 12 W

C 24 W

D 48 W

Your answer ☐ **[1]**

3. One million electrons travel between two points in a circuit.
The **total** energy gained by the electrons is 1.6×10^{-10} J.

What is the potential difference between the two points?

A 1.6×10^{-16} V

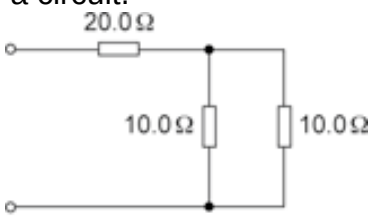
B 1.6×10^{-4} V

C 1.0×10^{3} V

D 1.0×10^{9} V

Your answer ☐ **[1]**

4. Three resistors are connected in a circuit.



The resistance of each resistor is shown in the circuit diagram.

What is the total resistance of this circuit?

A 10.0 Ω

B 20.2 Ω

C 25.0 Ω

D 40.0 Ω

Your answer ☐ **[1]**

5. A small heater is connected to a power supply. The power supply is switched on for 100 s. The current in the heater is 3.0 A and it dissipates 1200 J of thermal energy.

What is the potential difference across the heater?

A 0.25V

B 4.0V

C 12V

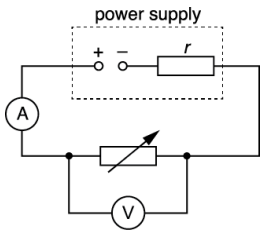
D 300V

Your answer ☐

[1]

6. A variable resistor is connected across the terminals of a power supply of constant e.m.f. and internal resistance *r*.

© OCR 2017. You may photocopy this page.



The resistance of the variable resistor is changed from zero to its maximum value.

Which of the following statements is/are correct?

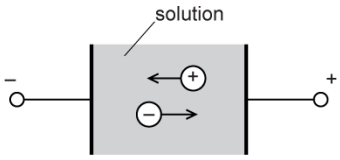
1 The current in the circuit decreases.
2 The p.d. across the internal resistance decreases.
3 A graph plotted of terminal p.d. against current has a negative gradient.

A Only 1
B Only 1 and 2
C Only 1 and 3
D 1, 2 and 3

Your answer ☐

[1]

7. The diagram below shows the motion of positive and negative particles in a conducting solution.

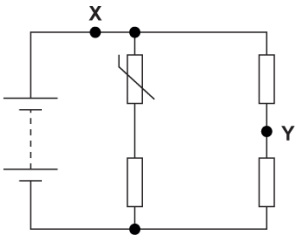


Which statement is correct?

A The current in the solution is zero.
B The conventional current is to the left.
C The positive particles are always protons.
D The negative particles are always electrons.

Your answer ☐ **[1]**

8. A circuit is shown below.



The battery has negligible internal resistance. The temperature of the NTC thermistor is **decreased**.

Which of the following statements is / are correct?

1 The current at **X** increases.
2 The current at **Y** remains the same.
3 The potential difference across the thermistor increases.

A 1, 2 and 3
B Only 2 and 3
C Only 3
D Only 2

Your answer ☐ **[1]**

9. Which law indicates that charge is conserved?

A Lenz's law
B Coulomb's law
C Kirchhoff's first law
D Faraday's law of electromagnetic induction

Your answer ☐ **[1]**

10(a). **Fig. 17.1** shows a resistor and a diode connected in series to a cell.

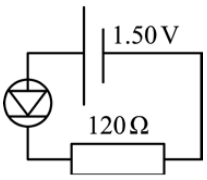


Fig. 17.1

The resistor has resistance 120 Ω. The cell has e.m.f. 1.50 V and negligible internal resistance. The potential difference across the diode is 0.62 V.

Calculate the total power dissipated in the circuit.

power = ... W **[3]**

© OCR 2017. You may photocopy this page.

Created in ExamBuilder

(b). A student designs a circuit to vary the brightness of a filament lamp. The circuit is shown in **Fig. 17.2**.

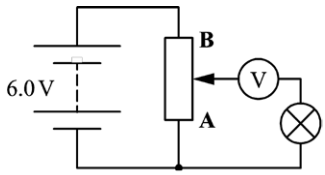


Fig. 17.2

The circuit is set up. Moving the slider from **A** to **B** changes the voltmeter reading from 0 V to 6.0 V but the lamp stays off. The lamp is not faulty.

Explain the observations above and refine the circuit design so that the brightness of the lamp can be varied as the slider is moved from **A** to **B**.

---**[3]**

11. A filament lamp **X** is part of an electrical circuit. The circuit has a battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance. The potential difference across the lamp can be increased **continuously** from 0 to 6.0 V. This potential difference is measured using a voltmeter.
The lamp glows brightly at 6.0 V.

i. Draw a circuit diagram for this electrical arrangement.

[2]

ii. Describe and explain the variation of the resistance of this lamp as the potential difference across it is changed from 0 to 6.0 V.

© OCR 2017. You may photocopy this page.

Created in ExamBuilder

.....

.....

.....

.....

..... **[4]**

iii. The filament lamp **X** is now connected in a different circuit as shown in Fig. 16.

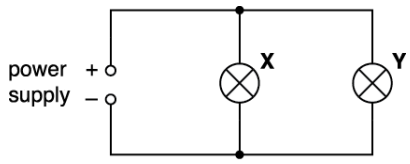


Fig. 16

The power dissipated in **X** is three times more than the power dissipated in the filamentlamp **Y**. The filament wire of lamp **X** has a diameter half that of lamp **Y**. The filament wires of **X** and **Y** are made of the same material and are at the same temperature.

Calculate the ratio

$$\frac{\text{mean drift velocity of charge carriers in lamp } \mathbf{X}}{\text{mean drift velocity of charge carriers in lamp } \mathbf{Y}}.$$

ratio = ... **[3]**

12(a).

Fig. 18.1 shows a circuit.

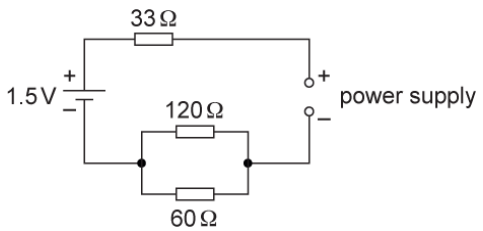


Fig. 18.1

The cell has e.m.f. 1.5 V. The cell and the variable power supply both have negligible internal resistance.

i. The e.m.f. of the power supply is set at 4.2 V.
Calculate the current *I* in the 33 Ω resistor.

I = ... A **[3]**

ii. The e.m.f. of the variable supply is now slowly decreased from 4.2 V to 0 V.
Describe the effect on the current *I* in the 33 Ω resistor.

---

[2]

(b). A group of students are investigating the power dissipated in a variable resistor connected across the terminals of a cell. The cell has e.m.f. 1.5 V.
The students determine the power *P* dissipated in the variable resistor of resistance *R*.

Fig. 18.2 shows the data points plotted by the students on a graph of *P* (*y*-axis) against

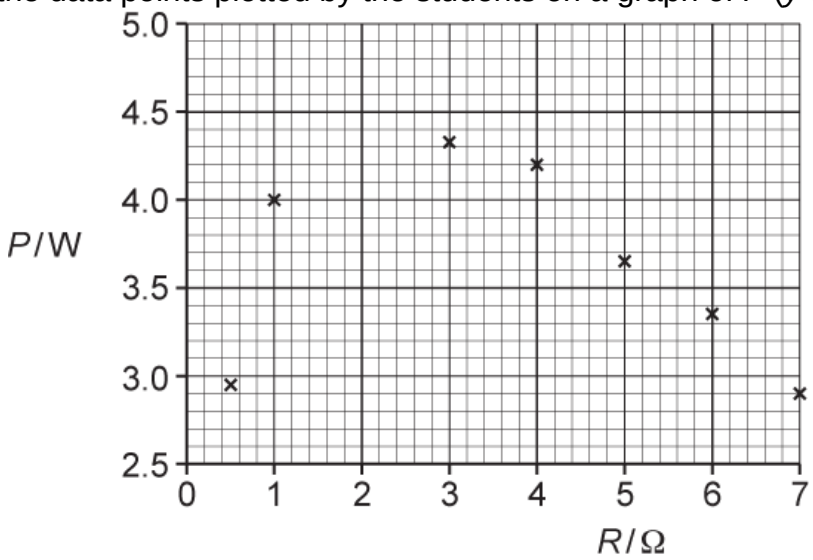


Fig. 18.2

The group of students know that **maximum power** is dissipated in the variable resistor when *R* is equal to the internal resistance *r* of the cell.

Describe, with the help of a suitable circuit diagram, how the students may have determined *P* and *R*.
Use Fig. 18.2 to estimate the internal resistance *r* of the cell and discuss any limitations of the data plotted by the group.

© OCR 2017. You may photocopy this page.

Created in ExamBuilder

[6]

13(a). Fig. 19.1 shows an electric circuit.

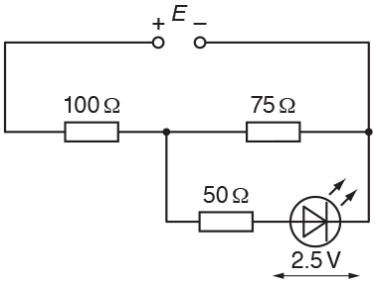


Fig. 19.1

The power supply has electromotive force (e.m.f.) *E* and negligible internal resistance.

The resistance values of the resistors are shown in Fig. 19.1. The *I*–V characteristic of the light emitting diode (LED) is shown in Fig. 19.2.

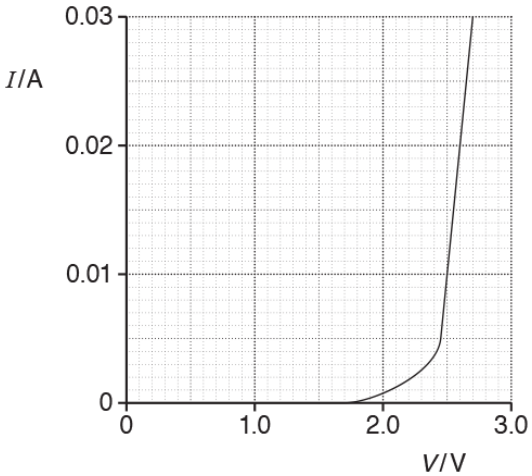


Fig. 19.2

The potential difference (p.d.) across the LED is 2.5 V.

Use Fig. 19.2 to show that the p.d. across the 50 Ω resistor is 0.50 V.

[2]

(b). Calculate the e.m.f. *E* of the power supply.

E = ...V **[3]**

END OF QUESTION PAPER

© OCR 2017. You may photocopy this page.

Created in ExamBuilder

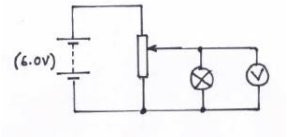
Mark scheme

Question			Answer/Indicative content	Marks	Guidance
1			B	1	**Examiner's Comments** This was a question on combining together three important expression in the topic of electricity; $V = IR$, $R = \rho L/A$ and $I = Anev$. On top of this, there was the additional information that **P** and **Q** were in parallel and hence the potential difference across each wire was the same. The mean drift velocity v of the electrons is given by the expression $v = \frac{V}{ne\rho L} \propto \frac{1}{L}$. The cross-sectional area A, and hence the diameter d of the wire has no effect on v. The relationship above implies that for wire **Q**, $v = \frac{1}{3} \times 0.60 = 0.20$ mm s^{-1}. The correct answer is **B**. All the distractors were equally popular. About a third of the candidates, mostly from the very top end of the ability range, were successful in this very demanding question.
			Total	**1**	
2			**A**	**1**	
			Total	**1**	
3			**C**	**1**	
			Total	**1**	
4			**C**	**1**	**Examiner's Comments** This was a well-answered question with most candidates demonstrating excellent knowledge of resistors in series and parallel combination. On many scripts, there was hardly any working shown. The two 10.0 Ω resistors in parallel gave a combined resistance of 5.0 Ω. This added to the series resistor of 20.0 Ω gives the correct answer of 25.0 Ω. The most popular distractor was **D** – where all the resistance values were simply added together.
			Total	**1**	
5			**B**	1	
			Total	**1**	
6			**D**	1	

© OCR 2017. You may photocopy this page.

Created in ExamBuilder

			Total	**1**	
7			**B**	**1**	**Examiner's Comment** All questions showed a positive discrimination, and the less able candidates could access the easier questions. The questions in Section A require careful reading and execution. Underlining or circling key information may help. Candidates are reminded not to use highlighter pens for this purpose. There is ample space for jotting down ideas and key equation, but it is best to do calculations on calculators to save time. Questions **1, 3,10** and **14** proved to be particularly straightforward, allowing most of the candidates to demonstrate their knowledge and understanding of physics. At the other end of the scale, Questions **5, 9,** and **15** proved to be more challenging. • Question **5** was on the superposition of waves and the relationship intensity∞ amplitude^2. The amplitude of the resultant wave is 0.4*a* and therefore the intensity of resultant wave must be 0.16*I*. The most popular distractors were **B** and **C**. Less than half of the candidates got the correct answer of **A**. • Question **9** was about doubling the separation between two oppositely charged parallel plates. The only correct statement is **D**. Electric field strength is p.d. divided by the distance between the plates. Since both quantities double, the electric field strength, must remain the same. • Question **15** was about refraction and the equation nsinη = constant at the boundary between two materials. The ratio *n*1/*n*2 = sin80°/sin90° = 0.98; the correct is **B**. The most popular distractor was **C**, which was the inverse of the correct answer.
•			**Total**	**1**	
8			**B**	**1**	
			Total	**1**	
9			c	1	**<u>Examiner's Comments</u>** This was a well-answered question with most candidates correctly recalling that charge is conserved according to Kirchhoff's first law. A significant number of candidates distracted towards **B**; perhaps because of the unit of charge is the coulomb.

			Total	**1**	
10	a		p.d. across resistor = 1.50 − 0.62 = 0.88 (V)	C1	
			current = 0.88 / 120 = 7.33… × 10^{-3} (A)	C1	
			power = *VI* = 1.50 × 7.33 × 10^{-3} = 1.1 × 10^{-2} (W)	A1	
	b		The voltmeter has large or infinite resistance.	B1	
			Hence the p.d. across the lamp or current in the lamp is small or zero (and the lamp is not lit).	B1	
			Refining design: remove voltmeter from the circuit or place the voltmeter across the lamp.	B1	
			Total	**6**	
11		i	Correct circuit with a battery, potential divider, lamp and voltmeter. 	B1	
		i	Correct symbols used for all components.	B1	**Allow**: A cell symbol for a battery
		ii	Description: The temperature of the filament increases. (AW)	B1	
		ii	The resistance of the lamp increases	M1	
		ii	from a non-zero value of resistance.	A1	**Allow** 'when cold the resistance is small'
		ii	Explanation: Resistance increases because electrons/charge carriers make frequent collisions with ions. (AW)	B1	
		iii	(*P* = *VI*) current in **X** is 3 times the current in **Y** Or area of **X** is 4 times smaller than area of **Y**	C1	**Allow** other correct methods.
		iii	*I* = *Anev* and ratio = $\frac{3}{0.25}$	C1	
		iii	ratio = 12	A1	
			Total	**9**	
12	a	i	Resistance of parallel combination = 40 (Ω)	**C1**	**Allow** $(1/60 + 1/120)^{-1}$
			$I = \frac{4.2-1.5}{40+33}$	**C1**	
			I = 0.037 (A)	**A1**	**Allow** 2 marks for $I = \frac{4.2+1.5}{40+33} = 0.078\,(\text{A})$
					Examiner's Comment

					The success in this question hinged on understanding the effect of two opposing e.m.f.s in a circuit and determining the total resistance of the circuit. About a third of the candidates produced well-structured and reasoned answer leading to the correct current of 0.037 A. Most candidates picked up a mark for determining the total resistance of the two parallel resistors (40 Ω). The total e.m.f. in the circuit is 2.7 V and the total resistance is 73 Ω. Those using a total e.m.f. of 5.7 V ended up with the incorrect current of 0.078 A; two marks were awarded for this answer. A small number of candidates tried to calculate the current using either using 1.5 V or 4.2 V or 33 Ω.
		ii	Any <u>two</u> from: The current decreases up to 1.5 V The current is zero at 1.5 V The current changes direction / is negative when < 1.5 V The current increases below 1.5 V	**B1×2**	**Allow** 'current is zero when the e.m.f.s are the same' **Examiner's Comment** Most of the answers here showed poor understanding of the circuit in Fig. 18.1. Nothing could be awarded for vague answers such as *'current decreases because $I \propto V$* or *'e.m.f. decreases so current decreases'.* The current decreases as the e.m.f. of the supply approaches 1.5 V, at 1.5 V the current is zero, the direction of the current reverses and its magnitude increases when the e.m.f. of the supply gets below 1.5 V. About a quarter of the candidates gave credible answers.
	b		**Level 3 (5-6 marks)** Clear description including a reasonable estimate of *r* **and** clear limitations *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3-4 marks)** Some description with an attempt to estimate *r* **and** some limitations *There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.* **Level 1 (1 -2 marks)** Limited description *There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.*	**B1×6**	Use level of response annotations in RM Assessor, e.g. L2 for 4 marks, L2^ for 3 marks, etc. **Indicative scientific points may include:** **Description and estimation** • Correct circuit with (variable) resistor, ammeter and voltmeter • Correct symbols used for all the components • *R* changed to get different values for *P* • $R = V / I$ (using ammeter and voltmeter readings) or *R* measured directly using an ohmmeter with the variable resistor isolated from the circuit or *R* read directly from a resistance box • Power calculated using $P = V^2/R$ or $P = VI$ or $P = I^2R$ • The value of *r* is between 1.0 to 3.0 Ω • A smooth curve drawn on Fig. 18.2 (to determine *r)* • A better approximation from sketched graph or *r* is between 1.5 and 2.7 Ω

			0 marks *No response or no response worthy of credit.*		• Any attempt at using $E = V + Ir$, with or without the power equation(s) to determine r - even if the value is incorrect **Limitations** • 'More data' required • Data point necessary at R = 2.0 Ω / More data (points) needed between 1 to 3 Ω • No evidence of averaging / Error bars necessary (for both P and R values) **Examiner's Comment** This was a level of response (LoR) question had three ingredients - drawing a viable circuit diagram that would enable the data shown in Fig.18.2 to be reproduced, using the figure to estimate the internal resistance of the cell and finally outlining any limitations of the data displayed in the figure. There is no one perfect model answer for a level of response question. A variety of good answers did score top marks. Most circuit diagrams were correct and well-drawn. There was the occasional mistake with the circuit symbol for a variable resistor; the thermistor symbol was a regular substitute. Most candidates drew a smooth curve on Fig. 18.2 and used this to estimate the internal resistance of the cell. Many also realised that the data points showed no evidence of averaging or error bars and that there were missing data points between 1.0 Ω and 3.0 Ω. Some candidates wanted *'more data points spaced regularly at interval of* *0.5* Ω', which was a sensible suggestion. Some weaker candidates attempted to draw a straight line of best-fit through the data points and then tried to determine the internal resistance from the gradient. There was a good spread of marks amongst the three levels.
			Total	**11**	
13	a		current = 0.01 (A) p.d. = 0.01 × 50 (= 0.50 V)	M1 A1	**<u>Examiner's Comments</u>** This was an accessible question on determining the p.d. across the LED using the data from **Fig. 19.2**. The universal approach was short and precise: V = 0.01 × 50 = 0.50 V. However, a significant number of candidates used a longer route involving the potential divider rule and the 250 Ω resistance of the LED.
	b				**Allow** other correct methods **Note** there is no ECF from (**a**)

			(V_{75} =) **0.5 + 2.5** (V) **or** (R_{LED}) = **250** (Ω) **or** (R_p =) **60** (Ω) (I_{100} =) **0.05** (A) (E = 3.0 + 0.05 × 100) E = 8.0 (V)	C1 C1 A1	**Allow** 1 SF for the p.d. of 3 (V) There is no ECF here from wrong physics (XP) from the parallel network **Allow** 1 SF answer of 8 **<u>Examiner's Comments</u>** The analysis of the circuit proved to be problematic with most of the candidates getting as far as calculating either the resistance of the LED as 250 Ω or the p.d. across the LED-50 Ω resistor combination as 3.0 V. The stages thereafter demonstrated all the usual misconceptions; these are summarised later. About a quarter of the candidates produced flawless solutions using a range of techniques from Kirchhoff's two laws to potential dividers. The simplest solution had the correct current of 0.050 A in the 100 Ω resistor, followed by the correct value of the e.m.f. of 8.0 V. This type of solution is shown in exemplar 7. **Misconception** These were the most common errors made in calculating the e.m.f. of the power supply. • Calculating the total resistance of the parallel network by omitting the resistance of the LED. • The current in the 100 Ω resistor was the same as the current of 0.010 A in the LED. • The current in the 100 Ω resistor was the same as the current of 0.040 A in the 75 Ω resistor. • Using the potential divider equation by completely omitting the LED-50 Ω resistor series network. **Exemplar 7** (b) Calculate the e.m.f. E of the power supply. $R = \frac{2.5}{0.01} = 250\Omega$ $50 + 250 = 300$ $\frac{300}{75} = 4$ $4 \times 0.01 = 0.04$ $0.04 + 0.01 = 0.05$ $100 \times 0.05 = 5V$ $0.5 + 2.5 + 5 = 8V$ E = 8 V [3] This exemplar shows a perfect response from a middle-grade candidate. The response should have been written to 2 SF. However, because the answer was 8.0 V, a 1 SF response was allowed without incurring any penalty.
			Total	**5**	